

Let's Encrypt HTTPS — Integration Status

Revision: 3 **Last modified:** 2026-07-01T11:42:00Z **Status:** In progress — Phases 0-3 + 5 PASS: a REAL hermetic DNS-01 certificate is issued by local Pebble AND auto-renewed/rotated (S1→S2, new serial, zero-downtime swap), both cert-analyzer-verified + re-runnable + guarded. All autonomous LE phases are DONE. Phase 4 (LE staging) + Phase 6 (production cutover) remain OPERATOR-BLOCKED (design §9 real-domain gate).

Authority: Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc for the Let's Encrypt HTTPS workstream (task #59). **Companion:** design ../LETSENCRYPT_HTTPS.md · plan ../LETSENCRYPT_HTTPS_PLAN.md · summary Status_Summary.md

Operator-blocked items (read first — §11.4.45 O(1) surface)

Item	Why blocked	Unblock condition
Phase 4 — LE staging end-to-end (real DNS-01)	Needs a real DNS provider + an API token stored as a Podman secret (§11.4.10 — value never in git/.env). Operator chose <i>defer real domain</i> (Phase 0).	Operator provides the DNS provider name + a scoped API token, and names a test hostname.
Phase 6 — production cutover	Real public domain, real DNS-01 for that domain, single operator-gated go-live (design §9).	Operator authorises the go-live with the real domain + token + host firewall opening.

Operator decisions captured (Phase 0 — §11.4.66)

Decision	Choice	Date
TLS client / terminator	Caddy auto-HTTPS (in-process ACME client + terminator; no cron/CI/hook renewal — §11.4.156)	2026-07-01
ACME challenge type	DNS-01 (works without inbound :80/:443 to the host)	2026-07-01
Rollout order	Hermetic (Pebble) + LE-staging first; defer real domain — prod is one later operator-gated cutover	2026-07-01

Phase status (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Validation	Evidence
0 — Operator decisions + scaffolding	Capture the 3 design §9 decisions; scaffold the plan	PASS	decisions table above; committed LETSENCRYPT_HTTP + _PLAN.md @ fddd25c
1 — Cert-analyzer (client/challenge-agnostic)	Pure offline analyzer over a resulting PEM (cert_not_expired / cert_days_remaining / cert_san_matches / cert_chain_roots_in / cert_renewal_due) + golden-good/golden-bad self-validation (§11.4.107(10))	PASS	selftest 37/37; guard GREEN+RED PASS; §1.1 mutation (SAN exact→substring) → guard restore byte-identical md5 a06e0f89...; qa-results/letsencrypt/cert-analyzer/results/regression/cert_analyzer_selfvalidati
1 — Custom Caddy image w/ DNS-01 provider module	Build a rootless Caddy image bundling the local dns.providers.challtestsrv module (§11.4.161)	PASS	build.sh built localhost/helix_proxy/caddy-challtestsrv:2.8.4; post-b anti-bluff check caddy list modules grep dns.providers.challtestsrv (a build ≠ module linked, §11.4.6); qa-results/letsencrypt/build/
2 — Compose service + Podman	Wire Caddy + CoreDNS (zone-determination fix) + secret NAME/PATH (no	PASS	compose.hermetic.yml (4 ser PEBBLE_VA_ALWAYS_VALID=0; secrets commented for hermetic; caddy loopback

Phase	Scope	Validation	Evidence
secret + cert volume + CoreDNS SOA front	value) + persistent cert volume		bound per security-audit M CoreDNS answers hermeti SOA, forwards _acme-chall TXT to challtestsrv
3 — Hermetic local-ACME with Pebble (anti-bluff core)	Caddy issues a REAL cert against local Pebble via DNS-01, fully offline; cert-analyzer asserts the served leaf	PASS	re-runnable phase3_hermetic_issue.sh — clean runs each issued a r cert (Pebble VA_ALWAYS_VALID=0 gen validation); cert-analyzer l not_expired + SAN proxy.hermetic.test + neg SAN-reject + chain-to-th run's-Pebble-CA ; qa-resu letsencrypt/phase3_issuanc 20260701T100119Z/ + ...T1004
4 — LE staging end-to-end (real DNS-01)	Real LE staging endpoint + real DNS-01	OPERATOR-BLOCKED	needs DNS provider + tok (\$11.4.10)
5 — Renewal + rotation simulation (zero-downtime)	Force the ARI renewal window → Caddy renews via the ACME renewal path → analyzer confirms the new cert; 0 dropped across the swap	PASS	re-runnable phase5_rotatio S1→S2 (new serial, later notBefore), caddy log certificate needs renewal on ARI window→renewed successfully, swap ok=6 fa cert-analyzer PASS (S2→p Pebble CA); guard phase5_rotation_guard.sh RED+GREEN proven (wir run-tests.sh); qa-results/ letsencrypt/phase5_rotatio Trigger mechanism (stora surgery on the cached AR restart) is source-justified docs/research/ caddy_2110_ari_refetch_202 — in production Caddy re on its own schedule, no tr needed
6 — Production cutover + rollback runbook	Real domain go-live	OPERATOR-BLOCKED	operator go-live gate (des §9)

Honest boundary (§11.4.6)

Phases 0-3 + 5 are DONE with captured evidence. Issuance (Phase 3): the custom Caddy image is built with the DNS-01 module linked (anti-bluff-verified), the hermetic stack is wired (CoreDNS SOA front fixes certmagic's zone-determination), and `phase3_hermetic_issue.sh` issues a REAL cert via DNS-01 against local Pebble (`PEBBLE_VA_ALWAYS_VALID=0`) that the cert-analyzer verifies over the served leaf (incl. chain to THIS RUN's Pebble CA). Renewal (Phase 5): `phase5_rotation.sh` forces the ARI renewal window and Caddy renews via the ACME renewal path (certificate needs renewal based on ARI window→renewed successfully) to a NEW serial with a later `notBefore`, the renewal SWAP measured zero-downtime (0 dropped), and the cert-analyzer verifies the new leaf. Both are re-runnable (§11.4.98/§11.4.107) + guarded (§11.4.135, RED+GREEN). Honest trigger boundary (§11.4.6): the hermetic renewal is forced via storage-surgery on the cached ARI window + a Caddy restart because certmagic v0.21.3 (and even Caddy $\geq 2.11.0$ — source-proven docs/research/caddy_2110_ari_refetch_20260701/) offers no on-demand ARI re-fetch; in PRODUCTION Caddy renews on its own maintenance schedule with no trigger + no restart. NOT exercised (honestly): the real-domain paths — Phase 4 (LE staging)

- Phase 6 (production cutover) — wait on the operator gate; never a metadata-only PASS (§11.4 / §11.4.1).